

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES

(Established by the Govt. of Andhra Pradesh and recognized as per Section 2(f), 12(B) of UGC Act, 1956)

Campuses: Nuzvid | Rajiv Knowledge Valley (RK Valley) | Srikakulam | Ongole

SYNCHRONIZED UNDERGRADUATE B.TECH CURRICULUM STRUCTURE

Common Unified Syllabus Framework across Nuzvid, RK Valley, Srikakulam & Ongole Campuses

1. PROGRAMME PHILOSOPHY & OBJECTIVES

The RGUKT synchronized B.Tech curriculum is designed to impart rigorous fundamental engineering principles, project-based experiential learning ('Learning by Doing'), industry exposure, and research training. All four constituent campuses follow identical semester course codes, laboratory protocols, and credit weights.

2. SEMESTER-WISE CREDIT SUMMARY (4-YEAR B.TECH)

Engineering Year	Semester	Theory Credits	Lab / Practical Credits	Total Semester Credits
E1 (First Year)	Semester 1	15	5	20
E1 (First Year)	Semester 2	16	4	20
E2 (Second Year)	Semester 3	16	5	21
E2 (Second Year)	Semester 4	15	6	21
E3 (Third Year)	Semester 5	16	4	20
E3 (Third Year)	Semester 6	16	4	20
E4 (Final Year)	Semester 7	12	8 (Major Project Part-I)	20
E4 (Final Year)	Semester 8	6	12 (Internship/Project-II)	18
TOTAL CREDITS	8 Semesters	112	48	160 Credits

3. CORE ENGINEERING DISCIPLINES & SPECIALIZATION TRACKS

- **Computer Science & Engineering (CSE):** Data Structures, Algorithms, OS, DBMS, Computer Networks, AI/ML, Cloud Computing, Full-Stack Development, Cyber Security.
- **Electronics & Communication Engineering (ECE):** Analog & Digital Circuits, Signal Processing, VLSI Design, Embedded Systems, IoT, Microwave & Optical Communications.
- **Mechanical Engineering (ME):** Thermodynamics, Fluid Mechanics, Kinematics, CAD/CAM, Robotics, Heat Transfer, Manufacturing Science, CFD.
- **Civil Engineering (CE):** Structural Analysis, Concrete Tech, Geotechnical Engineering, Transportation Systems, Water Resources, GIS & Remote Sensing.
- **Chemical & Metallurgical Engineering:** Mass Transfer, Reaction Engineering, Process Control, Materials Characterization, Extractive Metallurgy, Nanomaterials.

4. MOOCs, NPTEL & INTERNSHIP CREDIT TRANSFERS

- **NPTEL / SWAYAM Credit Transfer:** Students are permitted to complete up to **12 credits** of Professional and Open Electives through approved 8-week or 12-week NPTEL/SWAYAM MOOCs courses.
- **6-Month Industry Internship:** In E4 Semester 8, meritorious students with no active backlogs may undergo a full-semester industry internship at partnered companies.